

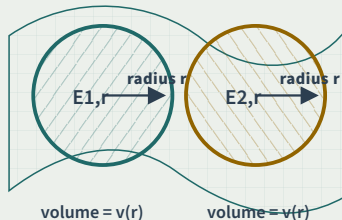
Why the common-radius volume terms cancel

and exactly where the argument can fail

STEP 1

One radius, compatible volumes

Both sources live in the same homogeneous measured space M ; the target lives in N .



target-ball volume = $w(r)$
Same radius;
a different dimension

STEP 2

Keep first order; control overlaps

$$P(E_i, r) = v(r) [f_{Si}(s_i) + o(1)]$$

$$P(Cr) = w(r) [f_T(t) + o(1)]$$

$$P(Cr \text{ intersection } E_i, r)$$

$$= w(r)v(r) [f_T, S_i + o(1)]$$

SOURCE OVERLAP

$$P(E_1, r \text{ intersection } E_2, r)$$

$$= O(v(r)^2)$$

= $o(v(r))$; therefore second order

TARGET-CONDITIONED TRIPLE OVERLAP

$$P(Cr \text{ intersection } E_1, r \text{ intersection } E_2, r)$$

$$= O(w(r)v(r)^2)$$

= $o(w(r)v(r))$; therefore second order

$$P(E_1, r \text{ union } E_2, r)$$

keeps the two $v(r)$ density terms

STEP 3

Cancel only justified scales

COMMON-RADIUS LIMIT

numerator $\approx$

$$w(r)v(r) [f_T, S_1 + f_T, S_2]$$

denominator $\approx$

$$w(r)v(r) f_T [f_{S_1} + f_{S_2}]$$

$w(r)v(r)$ cancels; density ratio remains

FIRST-ORDER OVERLAP TEST

If $P(E_1, r \text{ intersection } E_2, r)$

$$= \kappa v(r) + o(v(r)),$$

the overlap is first order, not negligible.

The denominator changes.

The displayed Ehrlich density ratio no longer follows.

Population lemma only - no kNN guarantee

ASSUMPTIONS DO THE WORK

Joint absolute continuity; pointwise-continuous density versions; positive limiting density sums;
one common shrinking radius; overlap little-o conditions (local essential bounds suffice).